# Sepehr Mahmoudian

SENIOR AI ENGINEER

Machine Learning • Knowledge Graphs • Multimodal & Embodied AI

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GITHUB github.com/sepahead  LINKEDIN linkedin.com/in/sepehr-mahmoudian

Berlin-Brandenburg • available from Jan 2027 • German citizen • EN fluent / DE good / Persian native


## PROFILE

Builds custom agent harnesses for domain-specific applications, knowledge graphs and Graph RAG; evaluates LLMs, VLMs, VLAs and world models with rigorous, evidence-first engineering.

LLMs / VLMs • Knowledge Graphs • Agentic Vision • Multimodal AI • Scene Reconstruction

CONTEXT ENGINEERING • DATA PROVENANCE • MODEL EVALUATION • REINFORCEMENT LEARNING

AI ENGINEERING

**10+ YEARS**

RESEARCH TO PRODUCTION

PYTHON • RUST • AI / ML

## PROFESSIONAL EXPERIENCE

### Independent AI/ML Engineer & PhD Candidate

2020.04 - Present

**Self-employed - PhD candidate at Technische Universität Darmstadt** | Berlin & Darmstadt, Germany

- Building biophysical/functional neural models, PID (Partial Information Decomposition), Vision-Language-Action systems, VLA evaluation and neural control
- Building custom AI agent harnesses for domain-specific applications; multimodal 3D perception and auditable visualization

### Senior AI Engineer

2026.01 - 2026.03

**Neural Frames UG** | Berlin & Hamburg, Germany

- Evaluated LLMs, VLMs and end-to-end pipelines, tracing runs and results with Langfuse
- Contributed to Python pipelines for model evaluation and inference

### AI Engineering Resident

2025.09 - 2026.01

**Merantix AI Campus (Hacker Room)** | Berlin, Germany

- Built generative-media systems with Gaussian splatting and diffusion models for spatiotemporal anchoring
- Implemented confidential-computing smart contracts in Rust
- Deployed product services and data infrastructure across Cloudflare Workers, D1, R2, Google Cloud Run and AWS EC2

### Senior AI Engineer

2024.07 - 2025.07

**Vision Labs UG** | Berlin & Hamburg, Germany

- Built workflows for prompt optimization, LLM evaluation and observability with DSPy and Langfuse
- Designed data and MLOps pipelines on Azure and Google Cloud with staging, automated tests and CI/CD

### Research Associate

2021.07 - 2024.07

**Freie Universität Berlin** | Berlin, Germany

- Scaled and optimized large neural-network simulations in C/C++ and Python
- Built retrieval-augmented generation (RAG) workflows for technical literature and evaluated AI-agent reliability
- Applied statistical and machine-learning analysis in Python; reviewed code and research manuscripts in an international team

### Research Associate
2019.07 - 2020.06

**Georg-August-Universität Göttingen** Göttingen, Germany

Neural networks, Partial Information Decomposition (PID), information theory and reproducible data workflows

### Research Associate
2017.03 - 2019.07

**Goethe-Universität Frankfurt** Frankfurt am Main, Germany · Simulation and statistical analysis of neural-network models

### Visiting Scientist
2017.03 - 2019.03

**Max Planck Institute for Dynamics and Self-Organization** Göttingen, Germany · Statistical methods for neural-network analysis

### Research Associate
2015.02 - 2017.03

**Forschungszentrum Jülich** Jülich, Germany · Neural-network simulations and reinforcement-learning experiments

### Software Developer
2010.09 - 2012.03

**Centre for Content Creation** Cyberjaya, Malaysia · Award-winning iPhone/iPad applications and digital products

## EDUCATION

### PhD Candidate (Dr. rer. nat.)
2020.04 - Present

**Technische Universität Darmstadt** Darmstadt, Germany

- New mathematical PID analyses and information-theoretic representation analysis
- Engram's domain-specific AI agent produces provenance-tracked neural-model candidates for closed loop neural network simulations, with bounded compilation and simulation checks, in reproducible computational-neuroscience workflows

### Master's by Research in Biosciences: Neuroscience
2012 - 2014

**University College London** London, United Kingdom

Computational Neuroscience • Graduated with Distinction (Top 10%, CGPA 4.0/4.0)

### Bachelor of Science (Honours) in Computer Science
2009 - 2012

**Anglia Ruskin University** Cambridge, United Kingdom

Graduated with First Class Honours (Top 1%, CGPA 4.0/4.0)

## SELECTED AWARDS & FUNDING

**EDTH / 2025**
Special Mention at the European Defense Tech Hackathon Berlin 2025; co-organized with the Bundeswehr Cyber Innovation Hub

**HUMAN BRAIN PROJECT**
Research funding

**NEAR HORIZON**
Confidential-computing grant

**UCL & SHEFFIELD**
Competitive academic scholarships

**IOS APPLICATIONS**
15 international product awards

SIGNATURE
Sepehr Mahmoudian

PLACE Berlin, Germany
DATE 30 August 2026

PUBLIC ENGINEERING PORTFOLIO
GITHUB Open-source projects
github.com/sepahead
HUGGING FACE Models & datasets
huggingface.co/torusprime

## SELECTED PROJECTS

### engram
★ 15 · RESEARCH PROTOTYPE

**PROVENANCE-CONSTRAINED NEURAL R&D**

Domain-specific AI agent turns qualified evidence into neural-model candidates, with bounded compilation and simulation checks.

Neural Networks · Knowledge Graphs · Closed-loop Sim

### pid-rs
★ 14 · V0.9 SOURCE PRERELEASE

**VERIFIABLE INFORMATION DECOMPOSITION**

Measures how much information several signals share about an outcome, with built-in checks and provenance that keep results reproducible.

Rust / Python · Information Theory · Statistical ML

### prisoma
★ 2 · V0.9 PRERELEASE / ACTIVE R&D

**AUDITABLE CLOSED-LOOP EXPERIMENTS**

Simulation environment for closed-loop VLA and world-model experiments, with capture, replay and provenance.

Rust / Python · VLA + World Models · Embodied Sim

### NCP
★ 16 · V0.8 RELEASE / 1.0 RC BLOCKED

**NEURAL CONTROL PROTOCOL**

Links neural simulators, AI controllers and robots through four typed planes with local fail-safes and provenance.

Rust · Typed Control Planes · Zenoh

### manwe
★ 3 · ALPHA

**AIRSPACE PERCEPTION**

Research workbench for vision, acoustics, multi-camera geometry and tracking; model exports must pass artifact-identity and numerical gates.

Python · PyTorch · Multimodal Perception

### crebain
★ 21 · RESEARCH PROTOTYPE

**SPATIAL SITUATIONAL AWARENESS**

Unifies Gaussian-splat scenes, native object detection and multimodal 3D tracking in one tactical view for simulated drone operations.

Rust · Sensor Fusion · Native Inference

### melkor
★ 13 · HARDENING

**GAUSSIAN-SPLAT INTEGRITY**

Inspects and deterministically converts 3D Gaussian splats, surfacing numeric hazards and field provenance before pipeline use.

C++ · Scene Reconstruction · Asset Validation

### cobot-atlas
### relief-atlas
★ 7 · ⬇ 1,867 · PUBLISHED + DOI

**AI-GENERATED SIMULATION DATASETS**

2,023 robot-simulation and 10,079 disaster-relief GLB meshes with generation pipelines.

Python · Hugging Face · 3D / VLA

### galadriel
★ 7 · RESEARCH PROTOTYPE

**CROSS-SENSOR CONSISTENCY**

Fail-closed monitor for sensor disagreement on one track, using per-channel innovation tests and signed correlation over producer-declared projections.

Rust · NIS / CUSUM · Signed Correlation

### cortexel
★ 4 · 0.10.0-DEV.0

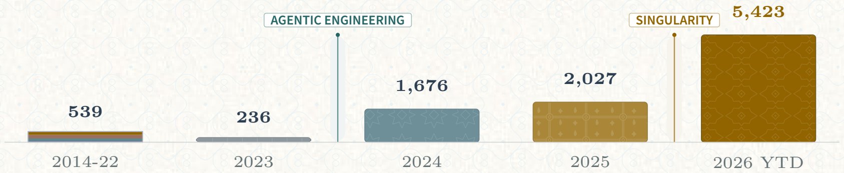
**AUDITABLE NEURAL FIGURES**

Checks scientific figures against declared contracts before rendering, so charts provably match their data and stay reproducible.

Figure Contracts · Deterministic SVG · Provenance

## PORTFOLIO & ENGINEERING ACTIVITY

| 9,901 | 23 | 112 | 146 |
|---|---|---|---|
| GITHUB CONTRIBUTIONS SINCE 2014 | NON-FORK PUBLIC REPOSITORIES | STARS ON PUBLIC REPOSITORIES | ACTIVE DAYS IN 2026 |




**CURRENT MOMENTUM**

**2.68×** 2026 YTD contribution volume versus the full 2025 calendar year

5,423 YTD • 2,027 in 2025

**CONTRIBUTION BY WEEKDAY**


**5,423** contributions • 2026 YTD

| | |
|---|---|
| Mon 16.8% | Tue 13.4% |
| Wed 11.8% | Thu 9.5% |
| Fri 11.8% | Sat 18.3% |
| Sun 18.4% | |

## TECHNICAL SKILLS

| | |
|---|---|
| • AGENTIC AI | LLMs / VLMs \| Agent Harnesses \| AI Agents \| Tool Use \| Model Evaluation \| Inference Pipelines |
| • MACHINE LEARNING | Python \| PyTorch \| Agentic Vision \| Multimodal AI \| Vision-Language-Action / VLA Evaluation |
| • KNOWLEDGE & RETRIEVAL | Knowledge Graphs \| Graph RAG \| RAG \| Embeddings \| Vector Search \| Reranking \| Data Provenance |
| • SYSTEMS & MLOPS | Rust \| C++ \| FastAPI \| ONNX \| TensorRT \| CUDA \| Metal / Core ML \| Docker |
| • CLOUD & INFRASTRUCTURE | GCP \| AWS \| Azure \| Kubernetes \| Terraform \| GitHub Actions \| Linux \| Zenoh |